

Chapter 19

The Problem of Adverse Selection

Learning Objectives

- 19.1 Explain insurance and how it works as a wealth-creating transaction.
- 19.2 Define and anticipate adverse selection.
- 19.3 Provide solution to the problem of adverse selection.
- 19.4 Explain screening and signaling.
- 19.5 Explain how online auction and sale sites address adverse selection.

Chapter 19 – Main Points

- Insurance is a wealth-creating transaction that moves risk from those who don't want it to those who are willing to bear it for a fee.
- Adverse selection is a problem that arises from information asymmetry—anticipate it, and, if you can, figure out how to consummate the unconsummated wealth-creating transaction (e.g., between a low-risk customer and an insurance company).
- The adverse selection problem disappears if the asymmetry of information disappears.
- Screening is an uninformed party's effort to learn the information that the more informed party has. Successful screens have the characteristic that it is unprofitable for bad “types” to mimic the behavior of good types.
- Signaling is an informed party's effort to communicate her information to the less informed party. Every successful screen can also be used as a signal.
- Online auction and sales sites, like eBay, address the adverse selection problem with authentication and escrow services, insurance, and on-line reputations.

Warm-Up

- Have you ever been nervous to buy something secondhand like a car, apartment, or phone because you weren't sure if the seller was hiding something?
- Why did you trust (or not trust) the seller, and how did that affect the price or your decision to buy?

Homeowners Insurance

- A homeowner pays \$1,200 per year for insurance that covers a potential \$300,000 loss from fire, theft, or natural disaster.
- Even though the home may never be damaged, the homeowner is willing to pay the premium to avoid the risk of financial ruin.
- Meanwhile, the insurer profits across a broad pool of low-probability risk holders.

Why it's wealth-creating:

- The homeowner values certainty and peace of mind more than the premium cost.
 - The insurer earns a return for bearing risk across many customers.
 - Both parties are better off, even though no wealth was physically created. It's created through improved utility and risk allocation.
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- One of the biggest challenges in transactions is risk — not knowing exactly what will happen, or what the other party knows.

Risk

- Risk is at the heart of hidden information problems. It explains why insurance exists, and why markets can fail without solutions like signaling and screening.
- We classify individual risk into three categories:
 - Risk aversion:
 - individuals are more likely to purchase insurance and more of it.
 - individuals will take less than an expected value to get the “sure thing”
 - Risk loving
 - individuals are less likely to purchase insurance
 - individuals are more likely to take the gamble than take the sure thing
 - Risk neutral
 - individuals are equally likely to purchase insurance
 - individuals will take the expected value of the payout

Discussion Activity 1

- Why is insurance considered a “wealth-creating transaction” even when no one gains money directly?
- Think in terms of risk allocation and utility. Who bears risk better? Why is that valuable?

Hidden Information – Adverse Selection

- **Adverse selection** arises from information asymmetry or hidden information
- It is when one party to a transaction is better informed than the other party
 - It will occur before the transaction takes place.
- Adverse selection makes it difficult to generate wealth creating transactions
- Examples
 - Insurance – insuring different parties who view risk differently
 - Used car sales – the market for lemons
- Differentiating between risk adverse, neutral, and loving individuals
- Differentiating between degrees of risk adverse individuals

Activity – Identifying Adverse Selection

Which of the following is not an example of adverse selection?

- A business bets the proceeds of a bank loan on the next NFL game.
- An accident-prone driver buys auto insurance.
- A patient suffering from a terminal disease buys life insurance.
- A really hungry person decides to go to the all-you-can-eat buffet for dinner.

Risk Aversion and Anticipation of Adverse Selection

- When an insurance company is trying to sell two policies to two individuals, but both have different expected costs, the company faces the adverse selection problem.
 - If it sells the policy for an average amount (the average between the low and high value customer), then it will lose the lower value customer as the lower value customer will take the chance on the expected cost of the risk.
 - The insurance company also loses money on the higher value customer, as the higher value customer would have paid a greater price.

Risk Aversion and Anticipation of Adverse Selection

- The insurance company, therefore, needs to anticipate adverse selection.
- The result is that the company may still lose out on low value customers, but it can charge a higher price to the high value customers.
- This is a price discrimination model, but we need to identify which customers are at a lower and higher risk.

Illustration of Anticipation and Risk

- An all-you-can-eat buffet attracts two types of customers. Regular customers value the buffet at \$20 and eat \$5 of food in costs to the restaurant. Hungry customers value the buffet at \$40 and eat \$10 of food. If there are 100 of each type in the market for a buffet dinner, what is the restaurant's profit?

$$[(20 \times 100) + (20 \times 100)] - [(5 \times 100) + (10 \times 100)] =$$

$$(2000 + 2000) - (500 + 1000) = \$2,500$$

$$30 \times 100 - 10 \times 100 = 3000 - 1000 = \$2,000$$

$$40 \times 100 = 4000 - 1000 = \$3,000$$

Risk Aversion and Adverse Selection

- Adverse selection and risk aversion go hand-in-hand. Individuals that are risk averse will enter a transaction when adverse selection isn't anticipated.
- Recall, the risk averse individual values the lottery at less than the expected value, and when a transaction occurs the one who doesn't want the risk sells it to the one who does
- Industries may engage in more risk aversion to deal with adverse selection:
 - Finance – Hedging strategies against potential losses
 - Healthcare – Preventive care and malpractice insurance
 - Retail – loss prevention (such as security and employee training) and price guarantees to attract consumers

Activity – Risk Aversion and Adverse Selection

Which of the following is not an example of risk aversion and adverse selection.

- A person with a family history of illnesses opts for a comprehensive health insurance plan that covers all major treatments and medications.
- A car insurance company offers lower premiums to drivers who have no history of accidents or traffic violations.
- A health-conscious individual decides to buy only organic food products, despite their higher cost.
- An individual in excellent health chooses a high-deductible health insurance plan because they anticipate low medical costs.

Activity – Adverse Selection

Which of the following is an example of adverse selection?

- A safe driver taking greater risk in a rental car than his own car.
- A terminally ill person purchasing life insurance.
- An employment contract encourages little effort on the part of employees.
- All the above

Designing Contracts to Combat Adverse Selection

- Good contract design helps insurers separate high-risk from low-risk customers by letting people reveal themselves through choice.
- Examples:
 - Health insurance: Bronze/Silver/Gold tiers in ACA marketplaces
 - Car insurance: High-deductible plans for safe drivers
 - Life insurance: No medical exam = higher premium
 - Pet insurance: No coverage for pre-existing conditions + 6-month waiting period for surgery

Designing Contracts to Combat Adverse Selection

Feature	Description	Impact
Co-pay	Customers pay a portion of the claim cost.	Discourages unnecessary or low-value claims.
Deductible	Customer must pay a minimum before coverage kicks in.	Screens out high-frequency claimants.
Waiting Period	Benefits delayed after enrollment (e.g., 90 days).	Discourages those seeking short-term coverage for known expenses.
Tiered Plans	Different coverage/premium levels based on needs.	Encourages self-selection: healthy people opt for basic plans, high-risk for full plans.

Discussion Activity 2

- How do companies use “menus” (like bronze, silver, gold plans) to let customers sort themselves?
- Consider how different customers reveal their risk levels or preferences when forced to choose.

Solving the Adverse Selection Problem

- We saw that insurance companies (and others) face the adverse selection problem, and that they need to anticipate it. We can extend this analysis now by looking at the following:
 - **Screening**
 - The efforts of the less informed party to gather more information about the more informed party.
 - Insurance companies use screening techniques in the form of deductibles, examining claim history and pre-existing conditions
 - Online dating services help eliminate adverse selection
 - **Signaling**
 - The efforts of the more informed parties to reveal information about themselves to the less informed party
 - College graduates use their degree as a signal to the market that they are a quality candidate
 - Warranties and guarantees also act as signals
- Both screening and signaling help insurance companies (and others) overcome the adverse selection problem.

Screening in Industries

- Health Care: Insurers often require potential policyholders to undergo health screenings or provide detailed medical histories before finalizing the terms of health insurance coverage.
- Finance: Banks and financial institutions use credit checks as a screening tool to assess the creditworthiness of borrowers.
 - Reviewing credit history, repayment patterns, and current debt levels, banks can make more informed decisions about loan approval and interest rates.
- Employment: Employers screen potential employees by checking references, conducting background checks, and requiring specific qualifications or certifications.
 - This process helps ensure that candidates are suitable for their roles and are likely to perform well.

Signaling in Industries

- Education: Individuals invest in higher education and obtain degrees or certifications, which act as signals to potential employers about their skills and commitment to their field.
 - An MBA degree is often regarded as a signal of advanced business management skills.
- Automotive: Car manufacturers might offer extended warranties on new vehicles as a signal of product quality and reliability.
 - This reassures potential buyers about the durability and maintenance costs of the vehicle, encouraging them to proceed with the purchase.
- Real Estate: Home sellers may hire independent inspectors to assess the condition of their property and share the results with potential buyers.
 - This signal of transparency and assurance about the home's condition can help attract buyers and justify the asking price

Screening vs. Signaling

	Screening	Signaling
Who initiates?	The less informed party	The more informed party
Purpose	To uncover private info from the other party	To credibly reveal private info to the other side
Common users	Insurers, employers, lenders	Job seekers, sellers, applicants
Typical tools	Tests, interviews, deductibles, credit checks	Degrees, warranties, certifications, branding
Economic effect	Reduces hidden information risk through vetting	Builds trust by offering costly, verifiable signals
Real-world example	Health insurer requires a physical exam	Applicant earns an MBA to signal competence

Activity – Identifying a Screen

Which of the following is an example of a screen?

- When someone is applying for a job, the employer checks references to determine the previous work habits of the applicant.
- When an employee purchases group life insurance without taking a physical exam, she knows more about her health than does the insurance company.
- When someone is considering buying a used car from a dealership, the seller knows the repair history of the car, but the potential buyer does not.
- A man buys an expensive birthday present for his girlfriend
- An insurance company offers a policy with a high deductible
- The seller of a used motorcycle knows more about its true condition than a prospective buyer
- Society supports long prison terms for corporate criminals

Activity – Identifying the Signal

Which of the following can be an example of a signal?

- An air-conditioning manufacturer offers a 50-year warranty.
- A lawyer offers to be paid only if the client wins.
- A student pursues an MBA.
- All the above

Activity – Design a Screening or Signaling Strategy

Scenario:

- You're helping a new online platform for luxury watches avoid fraud and misinformation, especially from counterfeit or damaged products being listed.

Instructions:

1. Design one screening tool the platform could use to verify sellers or products before a listing goes live.
2. Design one signaling tool the seller could voluntarily use to convince buyers of authenticity or quality.
3. Explain why your solution works to overcome adverse selection: how does it reduce information asymmetry?

Activity – Design a Screening or Signaling Strategy

- Screening: Require third-party verification or certification of authenticity (e.g., digital authentication)
 - Reduces bad actors from accessing the platform
- Signaling: Sellers offer a money-back guarantee or allow escrow with inspection period
 - Only confident sellers will offer this; reassures buyers
- Combined: Platform adds visible “trusted seller” badge based on transaction history and dispute ratio
 - Builds a feedback loop of trust and credibility

Discussion Activity 3

- What's the difference between a *screen* and a *signal*, and who uses each?
- Think about who knows more: the buyer or the seller? Who is trying to uncover vs. show info?

Case Study: Lemon Laws and Used Car Markets

- In the 1970s, economist George Akerlof described the “market for lemons”, used cars where sellers know more about car quality than buyers.
- Without transparency, buyers assume cars are bad and offer low prices.
- This drives good sellers out, leading to market collapse — a classic adverse selection case.

Case Study: Lemon Laws and Used Car Markets

Problem	Solution
Hidden defects	Required vehicle history reports (Carfax)
Risk of fraud	State “lemon laws” give buyers right to return faulty cars
Reputation risk	Online reviews (CarMax, Carvana, etc.)
Warranty signaling	Dealers offer 30–90 day warranties as signals of quality

Why do third-party services (like Carfax or certified programs) help both buyers *and* sellers?

Key Takeaways

- Insurance is wealth-creating when it reallocates risk to those who can bear it better.
- Adverse selection arises when one party knows more before the transaction, and that info skews the market.
- Hidden information causes high-risk individuals to pool with low-risk ones, raising costs or causing market breakdowns.
- Solutions include:
 - Screening by the uninformed party (e.g., applications, exams, history)
 - Signaling by the informed party (e.g., warranties, degrees, upfront info)
- Good contract design encourages self-selection and honest disclosure (e.g., bronze/gold plans).
- Online markets solve adverse selection through reputation, authentication, and transparency tools.
- Without intervention, adverse selection can lead to unconsummated wealth-creating transactions, deals that could benefit both sides but never happen.

Knowledge Check

1. Why is insurance a wealth-creating transaction?
2. What makes adverse selection a problem for insurers?
3. What's the difference between screening and signaling?
4. How does eBay reduce adverse selection?
5. Which of these is not an example of adverse selection:
 - a) Sick patient buys life insurance
 - b) Hungry person chooses buffet
 - c) Bad driver buys car insurance
 - d) Terminally ill person buys health insurance

Knowledge Check Answers

1. It reallocates risk to those who can bear it better, increasing utility.
2. Because high-risk individuals are more likely to buy, driving up average costs.
3. Screening = less informed party gathers info; signaling = more informed party reveals info.
4. Through reviews, ratings, escrow, and identity verification.
5. b) Hungry person choosing buffet is preference, not hidden information.

Consultant's Challenge: Solving Hidden Information Problems

Scenario:

- You've been hired by an auto insurance company to reduce losses from adverse selection. Too many high-risk customers are buying their policies, and low-risk customers are leaving due to rising prices.

Your Task:

1. How would you design contracts or features that encourage self-selection between high- and low-risk customers?
2. What screening or signaling mechanisms would you implement to better sort risk types up front?
3. How could you incorporate tools like reputation systems, data analytics, or premium structures to support long-term risk management?